

MIS 3033 – Computer-Based Information Systems

Section 001 | Fall 2026

Course Information

Instructor:	Adam Ackerman
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Email:	adam@ou.edu (include "MIS 3033" in subject line)
Location:	MFPH 3065
Meeting Times:	Tue/Thu 9:00 AM–10:15 AM
Course Site:	https://canvas.ou.edu
Final Exam:	Mon, Dec 14, 2026, 8:00 AM–10:00 AM, MFPH 3065

Prerequisites

MIS 3013 – Introduction to Programming

Students must complete MIS 3013 as a prerequisite because it provides essential foundational programming skills, which are necessary for applying object-oriented principles and developing desktop applications in this course. Additionally, MIS 3013 introduces Git, a version control tool that students will use to manage their code effectively throughout the data-driven project development process.

Textbooks & Software

[Required] Software: Visual Studio Community - Latest Version — <https://visualstudio.microsoft.com/downloads/>

[Recommended] Software: Git — <https://git-scm.com/>

[Optional] Software: Video recording software (zoom, kaltura, etc.) — <https://mymedia.ou.edu/>

Course Description

This intermediate-level course focuses on the design and development of data-driven desktop applications. Building on foundational programming skills, students will explore user interface design, adopt the Model-View-ViewModel (MVVM) architectural pattern (optional with bonus points), and integrate relational databases using both SQL and Entity Framework. Emphasis is placed on clean architecture, responsive user interfaces, and robust data handling. Through a mix of exams

and a final project, students will gain the experience and confidence to develop scalable, maintainable, and user-friendly applications.

Course Design

Skills in MIS 3033 are acquired through hands-on practice rather than rote memorization. Students will engage in repeated application of concepts in diverse contexts, fostering creative problem-solving. While some students may adapt quickly, others new to this approach may need time to develop this mindset—support from the professor and TAs is encouraged. Assessments will evaluate practical abilities (e.g., building UIs, managing databases) rather than memorized solutions, with exams and projects reflecting real-world challenges.

This course combines lectures, hands-on exercises, and discussions to build a foundation for future coursework and careers. Active participation is essential, and students may use AI tools for brainstorming or troubleshooting with proper citation.

Course Modules

Module 1: Application Development Fundamentals — *This module introduces the core principles of modern software development, including object-oriented programming, event-driven programming, and graphical user interface development. Students build desktop applications while developing proficiency with professional .NET development tools and practices.*

Module 2: Software Architecture & APIs — *This module focuses on designing maintainable software using architectural patterns and integrating applications through APIs and external data sources. Students learn how to create modular applications that communicate with other systems using modern software development practices.*

Module 3: Data Persistence & Enterprise Development — *This module explores data persistence, database integration, and enterprise application development using modern data access technologies. Students apply the concepts learned throughout the course by developing a comprehensive software application that solves a real-world business problem.*

Course Level Outcomes (CLOs)

CLO-1: Apply object-oriented programming principles to develop event-driven software applications. *[Assessed by: Homework 1, Homework 2, Homework 3, Exam 1, Group Project]*

CLO-2: Apply software architectural patterns and design principles to develop maintainable, modular, and scalable applications. *[Assessed by: Homework 3, Homework 6, Exam 2, Group Project]*

CLO-3: Design and implement data persistence solutions through relational databases and modern data access technologies. *[Assessed by: Homework 5, Exam 3, Group Project]*

CLO-4: Design, develop, and integrate application programming interfaces (APIs), external services, and other data sources to enable communication between software systems. *[Assessed by: Homework 2, Homework 4, Homework 6, Exam 2, Group Project]*

CLO-5: Develop a comprehensive software application that integrates user interface design, data management, software architecture, and system integration to solve a real-world business problem. *[Assessed by: Exam 3, Group Project]*

Grading & Assessments

Assessment Breakdown

Attendance – 10% (Sign-in sheet. Will drop two lowest scores (i.e. absences).)

Participation – 20% (Two lowest scores dropped)

Homework – 20% (Lowest score dropped)

Exams – 30% (Three exams all weighted the same)

Group Project – 20% (Presented during final exam time)

Attendance

Regular attendance is expected because course concepts are introduced through lectures, demonstrations, and hands-on activities that build upon one another throughout the semester. Students are responsible for all material covered in class, including announcements, demonstrations, and examples that may not be available outside of class.

Participation

For the design of this class to work, students must be actively involved in the class and attend the lecture. To ensure active learning, short surveys will be provided during class, typically consisting of a few questions, to gauge learning and ensure that my teaching is effectively received and understood by the class. The answers to the short surveys will be evaluated in class, and any concepts that are not understood will be explained. Students will receive full credit, no matter the answer, if they participate for the entire class period and submit it correctly.

Homework

Homework assignments are intended to reinforce course concepts and provide opportunities for both students and the instructor to identify areas requiring further attention. All assignments must be completed independently, though students may discuss general approaches with classmates,

TAs, or the instructor.

If a problem cannot be solved after substantial effort, students should note the specific difficulty in their submission and seek assistance from a TA. Submissions that contain identical wording, errors, or unusual similarities will be treated as academic dishonesty and addressed in accordance with the Honor Code. Everything that you submit you should be able to explain in order to receive your grade.

Exams

Each exam will consist of short problems like assigned homework and participation and will assess comprehension and application of course concepts. Exams are non-cumulative, each covering specific modules. Everything that you submit you should be able to explain to receive your grade.

Group Project

The semester-long group project provides students with the opportunity to apply the knowledge and skills developed throughout the course to solve a real-world business problem. Working in teams, students will design, develop, and present a comprehensive desktop application that integrates modern software architecture, database technologies, APIs, and user interface development. The project is completed in several phases to mirror the software development lifecycle used in industry.

Project Deliverables and Weights

The group project consists of the following deliverables, weighted as percentages of the overall project grade:

- **Data Preparation & Database Implementation: 25%**
- **Desktop Application: 35%**
- **API Development & System Integration: 20%**
- **Final Presentation: 20%**

Detailed instructions, requirements, and grading rubrics for each deliverable will be provided separately on Canvas.

Deliverable Descriptions

Data Preparation & Database Implementation (40%)

Teams will acquire or create a dataset, prepare and clean the data for use, design an appropriate relational database schema, and develop a utility application to create and populate the database using Entity Framework Core. This deliverable also includes documentation describing the dataset, database design, and any data preparation performed.

Desktop Application (40%)

Teams will develop a professional desktop application that connects to the database and provides users with the ability to view and manage data through a graphical user interface. The application should demonstrate sound object-oriented design, appropriate software architecture, input validation, and full CRUD (Create, Read, Update, Delete) functionality.

API Development & System Integration (5%)

Teams will modernize their application by developing a REST API that exposes application data and functionality. The desktop application will be updated to consume the API rather than communicating directly with the database, demonstrating a layered software architecture commonly used in modern software development.

Final Presentation (15%)

Each team will deliver a professional presentation demonstrating the completed application, database design, API architecture, and key technical decisions. Presentations should clearly communicate the project's purpose, implementation, challenges encountered, lessons learned, and each team member's contributions.

Team Evaluation and Individual Accountability

At the conclusion of the project, each student must complete a confidential team assessment evaluating both their own contributions and those of each teammate. The assessment will consider factors such as technical contribution, quality of work, communication, reliability, initiative, and overall professionalism.

Although the team assessment is **not assigned a separate point value**, completion of the

assessment is **required**. The results may be used to adjust individual project grades to more accurately reflect each student's contribution to the team's success. Students who do not meaningfully contribute to the project may receive a reduced project grade regardless of the overall quality of the team's final product.

Grading Scale

A: 90–100%

B: 80–89%

C: 70–79%

D: 60–69%

F: 0–59%

Course Policies

Academic Integrity

The University of Oklahoma has an Academic Misconduct Code that governs student academic performance in and out of the classroom. The Academic Misconduct Code can be reviewed online at <http://www.ou.edu/integrity>

Academic misconduct is defined as "any act that improperly affects the evaluation of a student's academic performance or achievement". All work assigned in this course is to be completed individually (unless otherwise stated). The steps and procedures as outlined in the Academic Misconduct Code will be followed in all cases of academic misconduct in this class.

MIS Division Guidelines for Academic Misconduct:

- All academic misconduct will be reported at: <http://www.ou.edu/content/dam/integrity/docs/academic-misconduct-report.pdf>
- The instructor will decide whether the misconduct is an Admonition or Violation of the Academic Integrity Code.
- When misconduct occurs on labs/similar assignments and is categorized as Admonition: a grade of zero will be recorded and a deduction equal to the value of the assignment will be applied.
- When misconduct occurs on an exam and is categorized as Admonition: a grade of zero will

be recorded.

- When misconduct occurs on an exam and is categorized as a Violation: no grade penalty is imposed until the student is officially found responsible. Until then, "N" is filed as the final course grade. If established, a violation results in a grade penalty and a University penalty.
- Faculty may impose greater grade penalties than those recommended above.

Laptop Requirements

A laptop computer is required for every class meeting. The following specifications are the minimum requirements:

Component	Minimum Specification
Processor	Intel Core i5 (12th gen or later), AMD Ryzen 5 5000 series or later, or Apple M1/M2
RAM	8 GB
Storage	512 GB SSD
Operating System	Windows 11
Required Software	Microsoft Office 365 (free via OU), SQL Server Management Studio (Windows) or Azure Data Studio (Mac)

Excused Absence Policy

Regular attendance is required. Absences will be considered excused only under the circumstances outlined below and only when appropriate documentation is submitted in a timely manner. The instructor retains sole discretion to determine whether an absence and its documentation meet the standards of this policy.

1. University-Recognized Excused Absences

A. Religious Observances

In accordance with University policy, absences for the observance of major religious holidays will be excused. Students must provide written notice as early as possible and no later than the end of the second week of the semester, or as soon as the conflict becomes known.

B. Official University-Sponsored Activities

Absences resulting from participation in official University-sponsored activities are excused. Written verification from the sponsoring University office is required. Advance notice is expected whenever practicable.

2. Medical Emergency

A medical emergency is a serious, acute condition requiring urgent or emergent medical treatment that renders the student unable to participate in academic activities.

Examples include, but are not limited to:

- Emergency room treatment
- Hospitalization
- Urgent surgical procedures
- Serious injury requiring immediate medical care

Required Documentation

Official documentation from a licensed healthcare provider or medical facility must include:

- Date(s) of treatment
- A statement confirming the student was medically unable to attend class on specified date(s)
- Provider signature or institutional verification

Documentation that merely confirms an appointment occurred, without stating incapacity to attend class, is insufficient.

Routine illness (e.g., cold, flu, minor viral illness, headache, fatigue, or similar conditions) does not qualify as a medical emergency and is not an excused absence under this policy, even with a doctor's note.

3. Motor Vehicle Accident

An absence resulting from a motor vehicle accident may be excused if the incident directly prevents class attendance.

Required Documentation

Students must provide a copy of the official police report documenting the accident. The report must clearly indicate the date and time of the incident and demonstrate that it coincided with the scheduled class time or otherwise made attendance not reasonably possible.

If medical treatment resulted from the accident, documentation meeting the requirements of the Medical Emergency section may also be required.

4. Family Emergency

A family emergency is a significant, unforeseen event involving an immediate family member that requires the student's presence.

Immediate family member is defined as: spouse or domestic partner; child; parent or legal guardian; sibling; grandparent; grandchild; or corresponding in-law relationship.

Qualifying circumstances may include:

- Death of an immediate family member
- Serious medical emergency involving an immediate family member

Required Documentation

Appropriate third-party verification must be provided (e.g., obituary, funeral documentation, hospital verification, or written confirmation from a responsible official). Verification through the Dean of Students Office may be required.

5. Court Appearance

An absence may be excused when a student is legally required to appear in court and the appearance directly conflicts with scheduled class time.

Excused court appearances may include, but are not limited to:

- Jury duty
- Subpoenaed testimony
- Required court hearings

Voluntary legal meetings, consultations with attorneys, or court appearances scheduled at the student's discretion do not qualify.

Required Documentation

Students must provide official court documentation that clearly indicates:

- The date and time of the required appearance
- Verification that the student's presence was mandatory

Advance notice is expected whenever practicable.

6. Non-Qualifying Absences

The following do not constitute excused absences:

- Routine/chronic illness
- Personal travel
- Employment conflicts
- Oversleeping
- Minor appointments
- Technical difficulties
- Failure to plan appropriately

7. Notification and Timing

Students must notify the instructor as soon as reasonably possible. Required documentation must be submitted within one week of returning to class unless otherwise approved in writing. Failure to provide timely notice or adequate documentation may result in the absence being recorded as unexcused.

Getting Help from TAs

Students are strongly advised to get help from the TAs when having problems with the class material. Their job is to help you when you get stuck, so feel free to approach them. They recently took this class and now really understand the problems you may run into.

Grade / Score Appeals

Students wishing to appeal a grade or score must submit the appeal to the instructor **in writing within 48 hours** of receiving the graded work. Score changes are at the discretion of the instructor, not the TAs.

It is important to understand that your score may go up or down based on a complete review of the work in question. It is usually the case that changing a few points on an assignment rarely makes a difference in the final grade.

Missed Exams / Make-Up Policy

Make-up exams will be given only in extraordinary circumstances. If you expect to miss an exam

or are unable to meet another requirement, please discuss this with the instructor **before the scheduled date, no later than 48 hours before the due date.**

Incomplete Grades

A grade of Incomplete may be awarded to students who have a legitimate reason for needing additional time to complete a course. Legitimate reasons include emergencies or extenuating circumstances that prevent a student from completing course requirements within the normal time frame.

Students must initiate the request for an incomplete **prior to the end of the semester**, and must be in good standing in the course. In no case will a grade of Incomplete be awarded to someone seeking more time to master the course material in order to improve their grade.

University Policies

Mental Health Support Services

As a university student, you may experience a range of issues that can cause barriers to learning, such as strained relationships, increased anxiety, alcohol/drug problems, feeling down, difficulty concentrating and/or lack of motivation. These mental health concerns or stressful events may diminish your academic performance and/or reduce your ability to participate in daily activities. OU offers services to assist you with addressing these and other concerns you may be experiencing. You can learn more about the broad range of confidential mental health services available on campus via the Counseling, Wellness, and Prevention Services (CWPS) at <https://www.ou.edu/health/cwps>, (405) 325-2911 or at Goddard Health Center, 620 Elm Avenue. Additionally, you can also access TimelyCare, a virtual health and well-being platform that provides 24/7/365 access to medical care, mental health support, and wellness services. To connect with TimelyCare, visit [timelycare.com/OU](<https://timelycare.com/OU>) or download the app.

Title IX / Sexual Misconduct

The University of Oklahoma is committed to providing a safe learning environment for all members of our community. Title IX of the Education Amendments of 1972 prohibits sex discrimination in educational programs and activities. This includes sexual harassment, sexual

assault, domestic violence, dating violence, and stalking.

As your instructor, I am a **mandatory reporter**. This means that if you share information about sexual misconduct or harassment with me — whether in person, in writing, in class discussions, on Canvas, or in any other setting — I am required to report that information to the OU Institutional Equity Office. If you need confidential support, you may contact **OU Advocates** at (405) 615-0013. You can also reach the Institutional Equity Office at (405) 325-3546 or visit <https://www.ou.edu/eoo>.

Reasonable Accommodation

The University of Oklahoma is committed to making its activities accessible to all participants. If you have a disability and require accommodations in this course, please contact the **Accessibility and Disability Resource Center (ADRC)** through iAdvise to schedule an appointment at <https://www.ou.edu/adrc>, by emailing adrc@ou.edu, or by calling (405) 325-3852. Accommodations are not retroactive. You must provide your accommodation letter to me as early as possible in the semester to ensure appropriate arrangements can be made.

Religious Observance

It is the policy of the University to excuse the absences of students that result from religious observances and to provide without penalty for the rescheduling of examinations and additional required classwork that may fall on religious holidays [Faculty Handbook 3.15.2]. If a religious holiday conflicts with a class meeting, quiz, exam, or assignment deadline, please notify me at least one week in advance so that we can make appropriate arrangements.

Pregnancy Adjustments

Title IX protects students from discrimination based on pregnancy, childbirth, or related conditions. Students who are pregnant or experiencing pregnancy-related conditions may request reasonable adjustments. For assistance, contact the **Institutional Equity Office** at (405) 325-3546 or visit <https://www.ou.edu/eoo>.

Final Exam Preparation Period

The 7 calendar days before the first day of the final exam period are designated as the final exam preparation period. During this time, no major homework, projects, presentations, or other graded work may be assigned that is due before the final exam period begins. Instructors may continue to cover new material during this period. Please refer to the Academic Calendar for the specific dates each semester.

Emergency Protocol — Severe Weather

In the event of severe weather, follow OU Alert instructions. If a tornado warning is issued while you are on campus:

- **Get In** — Move to the nearest interior shelter location.
- **Get Down** — Go to the lowest floor possible.
- **Cover Up** — Protect your head and neck.

Building-specific severe weather refuge maps are posted throughout campus. Sign up for **OU Alert** at <https://www.ou.edu/oupd/oualert> to receive emergency notifications.

Emergency Protocol — Active Threat (Run, Hide, Fight)

In the event of an active threat on campus:

- **Run** — If there is a safe escape path, leave immediately and call 911.
- **Hide** — If evacuation is not possible, find a secure location, lock/barricade the door, and silence your phone.
- **Fight** — As a last resort, use whatever means necessary to disrupt and disarm the threat.

OU Alerts are sent via the one.ou.edu system. Emergency contacts: OUPD Norman (405) 325-1717 / (405) 325-1911; OUPD Tulsa (918) 660-3900 / (918) 660-3333.

Emergency Protocol — Fire Alarm

In the event a fire alarm sounds:

1. **Leave** the building immediately.
2. **Know** your nearest exit route.
3. **Assist** those who need help evacuating, if it is safe to do so.
4. **Proceed** to the designated assembly area.
5. **Notify** emergency personnel of anyone who could not evacuate.
6. **Wait** for official clearance before re-entering the building.

Why You Belong at the University of Oklahoma

The University of Oklahoma is committed to creating a diverse and inclusive community where all students feel welcomed, valued, and supported. We recognize and respect the unique backgrounds, perspectives, and experiences that each student brings to our campus. Regardless of your race, ethnicity, gender identity, sexual orientation, religion, socioeconomic status, age, disability, or any other aspect of your identity, you belong here.

We strive to foster an environment that celebrates diversity and promotes understanding, empathy, and respect. We encourage you to embrace your own identity and to appreciate the richness that comes from engaging with others who are different from you. By learning from one another, we can collectively grow and create a more equitable and just world.

If you ever feel that you don't belong or experience any form of discrimination or harassment, please know that there are resources available to support you. You are not alone, and we are here to help you navigate any challenges you may encounter. Together, let's build a community where everyone can thrive and succeed.

Course Reflection Survey

Near the end of the semester, the university will administer an online course reflection survey. Your feedback helps improve instruction and course design. Surveys are confidential; individual responses are not released to instructors until after final grades have been submitted. I encourage you to take a few minutes to share your honest perspective.

Copyright Statement

Course materials prepared by the instructor are the intellectual property of the instructor. Students are expressly prohibited from selling notes or other course materials to any person or commercial firm without the express written permission of the instructor. Recording of lectures or class sessions is permitted only with the instructor's permission. Any approved recordings are for personal academic use only and may not be distributed, published, or posted publicly in any form.

Generative AI Policy

The use of generative AI tools is permitted in this course for the following purposes:

- Searching for and summarizing published academic or professional sources
- Creating outlines, draft text, or speech notes for presentations
- Receiving grammatical and stylistic feedback on your own drafts
- Further study and exploration of course topics outside of class
- Troubleshooting software errors (SQL, Python, Excel, etc.)
- Generating initial ideas for projects or assignments

Required disclosure: Any submission that includes AI-generated or AI-assisted content must include:

1. A declaration at the top of the document stating that AI was used.
2. An appendix containing the exact prompts submitted and the AI responses received.

Use of AI tools for purposes not listed above without proper acknowledgment and disclosure constitutes a violation of the OU Academic Integrity Code and will be handled accordingly.

Note: Regardless of the above, no AI tools of any kind are permitted during quizzes, exams, or any other timed in-class assessments unless the instructor explicitly states otherwise for a specific assessment.

Course Calendar

The schedule below is tentative; adjustments may be made. Announcements in class, via email, or on Canvas supersede

this outline.

Wk	Tue	Thu
1	<u>Tue, Aug 25, 2026</u> Course Overview + Environment Setup <i>GitHub, Visual Studio</i>	<u>Thu, Aug 27, 2026</u> 3013 Review / 3013 Review <i>Git Conditionals, Loops</i>
2	<u>Tue, Sep 1, 2026</u> 3013 Review <i>Functions, Classes</i>	<u>Thu, Sep 3, 2026</u> WPF <i>Lecture, In class example Participation</i> Homework 1
3	<u>Tue, Sep 8, 2026</u> WPF <i>Participation</i>	<u>Thu, Sep 10, 2026</u> File I/O + NuGet <i>Lecture, In class example</i> Homework 2
4	<u>Tue, Sep 15, 2026</u> File I/O + NuGet <i>Participation</i>	<u>Thu, Sep 17, 2026</u> File I/O + NuGet / File I/O + NuGet <i>Participation Participation</i> Homework 3
5	<u>Tue, Sep 22, 2026</u> Module 1 Review <i>Lecture, In class example</i>	<u>Thu, Sep 24, 2026</u> Exam 1
6	<u>Tue, Sep 29, 2026</u> JSON <i>Lecture, In class example</i>	<u>Thu, Oct 1, 2026</u> Consuming RESTful APIs <i>Participation Lecture, In class example; Group Formation</i>
7	<u>Tue, Oct 6, 2026</u> Consuming RESTful APIs <i>Participation</i>	<u>Thu, Oct 8, 2026</u> Consuming RESTful APIs / Project Work Day <i>Participation</i> Homework 4; Data for Project due and initial
8	<u>Tue, Oct 13, 2026</u> Consuming RESTful APIs	<u>Thu, Oct 15, 2026</u> Consuming RESTful APIs / Consuming RESTful APIs
9	<u>Tue, Oct 20, 2026</u> Entity Framework (EF) <i>Lecture, In class example</i>	<u>Thu, Oct 22, 2026</u> Entity Framework (EF) / Entity Framework (EF) <i>Participation Participation</i>
10	<u>Tue, Oct 27, 2026</u> Entity Framework (EF) <i>Participation</i> Homework 5	<u>Thu, Oct 29, 2026</u> Entity Framework (EF) / Project Workday <i>Participation</i>
11	<u>Tue, Nov 3, 2026</u> Module 2 Review	<u>Thu, Nov 5, 2026</u> Exam 2
12	<u>Tue, Nov 10, 2026</u> Creating RESTful APIs <i>Lecture, In class example</i>	<u>Thu, Nov 12, 2026</u> Creating RESTful APIs / Project Workday <i>Participation</i>

Wk	Tue	Thu
13	<u>Tue, Nov 17, 2026</u> Creating RESTful APIs <i>Participation</i>	<u>Thu, Nov 19, 2026</u> Creating RESTful APIs <i>Participation</i>
14	<u>Tue, Nov 24, 2026</u> Project Workday	<u>Thu, Nov 26, 2026</u> Thanksgiving Break <i>No Class</i>
15	<u>Tue, Dec 1, 2026</u> Module 3 Review <i>Homework 6</i>	<u>Thu, Dec 3, 2026</u> Exam 3 <i>Project due; Homework 7</i>
16	<u>Tue, Dec 8, 2026</u> Project Workday	<u>Thu, Dec 10, 2026</u> Project Workday
Final	FINAL EXAM Mon, Dec 14, 2026 · 8:00 AM – 10:00 AM · Room: MFPH 3065	